

Kumaresan M

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Backend-focused Full Stack Engineer with 2+ years of hands-on experience building production-grade web, cloud, and device-integrated systems. Strong expertise in Node.js, Python, AWS, and distributed backend systems, with real-world experience in healthcare, medical imaging (DICOM), IoT/edge devices, and e-commerce platforms. Proven ability to design scalable APIs, asynchronous processing pipelines, and reliable production systems.

CORE SKILLS

Backend: Node.js, Express.js, FeathersJS, Python, FastAPI (basic), REST APIs
Frontend: React.js, Next.js, Redux Toolkit
Database: PostgreSQL, MySQL, MongoDB, Firebase
Cloud DevOps: AWS (S3, SQS, Lambda, Secrets Manager), Docker (basic), Linux
Distributed Systems: Async processing, message queues, background jobs
Device Healthcare: Raspberry Pi, Bluetooth communication, DICOM workflows
Tools: Git, Postman, Swagger

RELATED EXPERIENCE

Full Stack Developer

Mar 2024 – Present

Rugas Technologies Private Limited

Bangalore, KA

- Developed and maintained **production backend and frontend features** for **KarmaTickets.ca**, a Canada-based e-commerce platform, supporting **authentication, user workflows, and order processing** using **Next.js, Node.js, and FeathersJS**.
- Designed and implemented **secure REST APIs** with **role-based access control (RBAC)**, improving reliability and maintainability of core business workflows.
- Contributed to **RADX.report**, a medical ultrasound reporting platform, integrating **React** frontend with **FeathersJS** backend and supporting **DICOM-based medical reporting workflows**.
- Designed and deployed a **scalable DICOM image ingestion pipeline** using **AWS S3, SQS, and Lambda**, enabling **secure, asynchronous processing** of medical images in production.
- Built a **Python-based Raspberry Pi edge service** enabling **Bluetooth-based device pairing**, secure cloud communication, **structured logging**, and remote diagnostics for medical devices.
- Implemented **FOTA (Firmware Over-The-Air)** update mechanisms to **remotely manage and deploy** Raspberry Pi application versions across production devices.
- Designed **structured logging and monitoring** to debug **device-level and backend production issues** in Linux-based environments.
- Integrated multiple **AWS services**, including **S3** for secure storage, **SQS** for asynchronous messaging, **Lambda** for background processing, and **Secrets Manager** for secure credential management.
- Collaborated closely with **QA and product teams** to troubleshoot production issues, reduce defects, and improve overall system reliability.

EDUCATION

PSG College of Technology

Coimbatore, TN

Bachelor of Technology, Textile Technology

July. 2015 – May 2019

CERTIFICATIONS

- Attained completion certificate for **DSA training in Java**, covering fundamental data structures, advanced algorithms with Back-end specialization in **Spring and Spring Boot**. [🔗 Certificate](#)
- Strong foundation in data structures and algorithms; practiced problem-solving across arrays, strings, hashing, and recursion.